

Paper 2.50 - Correction Surface Claim Contract

CQE-paper-02.50

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-13

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-02.50.md

Paper 2.50 - Correction Surface Claim Contract

Purpose

Paper 2.50 defines what counts as a valid claim about the Paper 2 correction surface. It is the contract that prevents a residue from being hidden, overpromoted, or imported without its receipt.

Admitted Paper 2 Claims

The following claims are admitted from Paper 2:

```
Rule30(L,C,R) = Rule90(L,R) xor (C and not R)
the correction fires exactly on (0,1,0) and (1,1,0)
the firing rows project to D4 coordinates (2,0) and (3,1)
nonzero correction is an obligation-bearing residue, not proof closure
```

Claim Requirements

Any later paper using the Paper 2 correction surface must state:

```
which LCR state is being used
which update route is being compared
whether the correction value is 0 or 1
which receipt proves the correction row
whether the row is closed, obligation, candidate, or downstream proof input
```

Linked Receipt

The minimum receipt link for Paper 2 is:

```
paper: CQE-paper-02
theorem: Correction surface decomposition
receipt: production/formal-papers/CQE-paper-02/correction_surface_receipt.json
status: pass
```

The receipt is sufficient for the finite Rule 30 / Rule 90 correction theorem. It is not sufficient by itself for full D4 triality, boundary repair, or physical interpretation claims.

Boundary Failures

The following are boundary failures:

```
counting nonzero residue as a closed proof
using a correction row without naming the source state
changing the firing set without rerunning the verifier
importing D4 coordinates while claiming Paper 2 proves full D4 triality
dropping the failed route that produced the residue
```

Boundary failures are not deleted. They become typed obligations for Paper 3, Paper 4, or later repair layers.

Hidden-Guess Contract

When hidden-guess diagnostics are enabled, the answer must be recorded before the receipt is revealed. The guess record should include:

```
prompt  
predicted firing set  
predicted proof status  
revealed receipt  
match/mismatch  
next obligation if mismatch
```

This keeps the diagnostic useful as training data without allowing the answer to leak into the prediction.

Conclusion

Paper 2.50 lets later papers import the correction surface honestly. It keeps the finite theorem available while preserving the distinction between residue, obligation, and proof.